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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), DECEMBER 2019

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS
(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{bmatrix}$ (3)
- 2 If 2 is an eigen value of $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, without using its characteristic equation, find the other eigen values. (3)
- 3 If $f(x,y) = xe^{-y} + 5y$ find the slope of $f(x,y)$ in the x-direction at (4,0). (3)
- 4 Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, where $z = e^x \sin y + e^y \cos x$ (3)
- 5 Find the mass of the square lamina with vertices (0,0) (1,0) (1,1) and (0,1) and density function $x^2 y$ (3)
- 6 Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates. (3)
- 7 Test the convergence of the series $\sum_{k=1}^\infty \frac{k}{2k+1}$ (3)
- 8 Check the convergence of $\sum_{k=1}^\infty \frac{1}{k^{k/2}}$ (3)
- 9 Find the Taylors series for $f(x) = \cos x$ about $x = \frac{\pi}{2}$ up to third degree terms. (3)
- 10 Find the Fourier half range sine series of $f(x) = e^x$ in $0 < x < 1$ (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Solve the system of equations by Gauss elimination method. (7)

$$x + 2y + 3z = 1$$

$$2x + 3y + 2z = 2$$

$$3x + 3y + 4z = 1$$

- b) Find the eigenvalues and eigenvectors of (7)

$$A = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

- 12 a) Find the values of λ and μ for which the system of equations (7)

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = \mu$$

has (i) no solution (ii) a unique solution and (iii) infinite solution

- b) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}. \text{ Also write the diagonal matrix.}$$

Module-II

- 13 a) Let f be a differentiable function of three variables and suppose that (7)

$$w = f(x-y, y-z, z-x), \text{ show that } \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

- b) Locate all relative extrema of $f(x, y) = 4xy - y^4 - x^4$ (7)

- 14 a) Find the local linear approximation L to the function $f(x, y) = \sqrt{x^2 + y^2}$ (7)

at the point $P(3, 4)$. Compare the error in approximating f by L at the point $Q(3.04, 3.98)$ with the distance PQ .

- b) The radius and height of a right circular cone are measured with errors of at most 1% and 4%, respectively. Use differentials to approximate the maximum percentage error in the calculated volume. (7)

Module-III

- 15 a) Evaluate $\iint_R y \, dx \, dy$ where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$. (7)
- b) Use double integral to find the area of the region enclosed between the parabola $y = \frac{x^2}{2}$ and the line $y = 2x$. (7)
- 16 a) Evaluate $\int_0^{\frac{y}{2}} \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$ by reversing the order of integration (7)
- b) Use triple integrals to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. (7)

Module-IV

- 17 a) Find the general term of the series $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{1 \cdot 3 \cdot 5 \cdot 7} + \dots$ and use the ratio test to show that the series converges. (7)
- b) Test whether the following series is absolutely convergent or conditionally convergent $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$ (7)
- 18 a) Test the convergence of $\frac{x}{1 \cdot 2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{3 \cdot 4} + \dots + \frac{x^k}{k(k+1)} + \dots$ (7)
- b) Test the convergence of the series $\sum_{k=1}^{\infty} \frac{(k+1)!}{4! k! 4^k}$ (7)

Module-V

- 19 a) Find the Fourier series of periodic function with period 2 which is given below $f(x) = \begin{cases} -x; & -1 \leq x \leq 0 \\ x; & 0 \leq x \leq 1 \end{cases}$. Hence prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ (7)
- b) Find the half range cosine series for $f(x) = \begin{cases} kx & 0 \leq x \leq L/2 \\ k(L-x) & L/2 \leq x \leq L \end{cases}$ (7)

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- a) Find the Fourier series of $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ x^2 & 0 < x < \pi \end{cases}$ (7)
- b) Obtain the Fourier series expansion for $f(x) = x^2, -\pi < x < \pi$. (7)

Final Scheme/ Answer Key for Valuation*Scheme of evaluation (marks in brackets) and answers of problems/key***APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2019****Course Code: MAT101****Course Name: LINEAR ALGEBRA AND CALCULUS****(2019-Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

$$1 \quad A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \dots (2), \text{ Rank of } A = 3 \dots (1) \quad (3)$$

$$2 \quad 2 + \lambda_2 + \lambda_3 = 11 \dots (1) \quad 2 \lambda_2 \lambda_3 = 36 \dots (1) \quad \lambda_2 = 3 \text{ and } \lambda_3 = 6 \dots (1) \quad (3)$$

$$3 \quad \text{Slope in the } x \text{ direction } f_x = e^{-y} \dots (2) \text{ slope at } (4, 0) = 1 \dots (1) \quad (3)$$

$$4 \quad \frac{\partial z}{\partial x} = e^x \sin y - e^y \sin x, \quad \frac{\partial^2 z}{\partial x^2} = e^x \sin y - e^y \cos x \dots (1) \quad (3)$$

$$\frac{\partial z}{\partial y} = e^x \cos y + e^y \cos x, \quad \frac{\partial^2 z}{\partial y^2} = -e^x \sin y + e^y \cos x \dots (1)$$

Conclusion:.....(1)

$$5 \quad \text{Mass} = \iint \delta(x, y) dx dy \dots (1) \quad \int_0^1 \int_0^1 x^2 y dx dy \dots (1) \quad 1/6 \dots (1) \quad (3)$$

$$6 \quad \int_{\theta=0}^{\frac{\pi}{2}} \int_{r=0}^{\infty} e^{-r^2} r dr d\theta \dots (1) - \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[e^{-t} \right]_0^{\infty} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \dots (1) \quad \frac{\pi}{4} \dots (1) \quad (3)$$

$$7 \quad \lim_{k \rightarrow \infty} U_k = \lim_{k \rightarrow \infty} \frac{1}{(2 + 1/k)} = 1/2 \neq 0 \dots (2) \text{ Divergent } \dots (1) \quad (3)$$

$$8 \quad \lim_{k \rightarrow \infty} (U_k)^{1/k} = \lim_{k \rightarrow \infty} \frac{1}{k^{1/2}} = 0 < 1 \dots (2) \text{ Convergent } \dots (1) \quad (3)$$

$$9 \quad f(x) = \cos x, f'(x) = -\sin x, f''(x) = -\cos x, f'''(x) = \sin x, f^{(4)}(x) = \cos x \dots (1) \quad (3)$$

$$f(\pi/2) = 0, f'(\pi/2) = -1, f''(\pi/2) = 0, f'''(\pi/2) = 1, f^{(4)}(\pi/2) = 0 \dots (1)$$

$$f(x) = \frac{(x - \pi/2)}{1!} (-1) + \frac{(x - \pi/2)^3}{3!} + \dots (1)$$

OR Alternate method

Formula $b_n = \frac{2n\pi}{1+\pi^2 n^2} (1 - (-1)^n e) \dots (1+1)$ $f(x) = \sum_{n=1}^{\infty} \frac{2n\pi}{1+\pi^2 n^2} (1 - (-1)^n e) \sin n\pi x \dots (1)$ (3)

PART B

Answer one full question from each module, each question carries 14 marks

11 a)

Module-I

$$[A:B] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 3 & 2 & 2 \\ 3 & 3 & 4 & 1 \end{bmatrix} \dots (1) \quad \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & 7 & -2 \end{bmatrix} \dots (3)$$

$x = -3/7, y = 8/7, z = -2/7 \dots (1+1+1)$

b) $\lambda^3 - 12\lambda^2 + 39\lambda - 28 = 0 \dots (2)\lambda = 1, 4, 7 \dots (2)$ eigen vectors $[2 \ -1 \ 2]^T, [-1 \ 2 \ 2]^T, [-2 \ -2 \ 1]^T \dots (3)$ (7)

12 a)

Augmented Matrix $\dots (1)$ Reducing $[A:B] \sim \begin{bmatrix} 2 & 3 & 5 & 9 \\ 0 & -15 & -39 & -47 \\ 0 & 0 & \lambda-5 & \mu-9 \end{bmatrix} \dots (3)$ No

solution when $\lambda=5$ and $\mu \neq 9 \dots (1)$

unique solution when $\lambda \neq 5$ and μ may have any value $\dots (1)$

infinite number of solutions when $\lambda=5$ and $\mu=9 \dots (1)$

b) characteristic equation $\lambda^3 - 12\lambda - 16 = 0 \dots (1)$ Getting $\lambda = -2, -2, 4 \dots (1)$ (7)

Eigen Vectors $[1, 0, -1]$ $[1, 1, 0]$ $[1, 1, 2] \dots (1+1+1)$

$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \dots (1)$ Diagonal matrix $D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \dots (1)$

Module-II

13 a) Put $r = x - y, s = y - z, t = z - x \dots (1)$ $\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r} - \frac{\partial v}{\partial t} \dots (2)$ $\frac{\partial v}{\partial y} = -\frac{\partial v}{\partial r} + \frac{\partial v}{\partial s} \dots (2)$ (7)

$\frac{\partial w}{\partial z} = -\frac{\partial w}{\partial s} + \frac{\partial w}{\partial t} \dots (2)$

b) $f_x = 4y - 4x^3, f_y = 4x - 4y^3 \dots (1)$ (7)

$f_{xx} = -12x^2, f_{yy} = -12y^2, f_{xy} = 4 \dots (1)$

$f_x = f_y = 0$. Critical points $(0, 0), (1, 1), (-1, -1) \dots (2)$

$(0, 0)$ saddle point, $\dots (1)$ $(1, 1), (-1, -1)$ point of maxima $\dots (1+1)$

$$14 \text{ a) } L(x, y) = 5 + \frac{3}{5}(x-3) + \frac{4}{5}(y-4) \quad \text{----- (3)}$$

$$L(Q) = 5.008 \dots \dots (1) \quad L(Q) - f(Q) = 0.00019, \dots \dots (1)$$

$$|PQ| = 0.045 \dots \dots (1) \quad \text{Error} = \frac{|L(Q) - f(Q)|}{|PQ|} = 0.0042 \dots \dots (1)$$

$$b) \quad V = \frac{1}{3} \pi r^2 h \dots \dots (2) \quad \log V = \log \frac{1}{3} \pi + 2 \log r + \log h \dots \dots (2)$$

$$\frac{dV}{V} \times 100 = 2 \frac{dr}{r} \times 100 + \frac{dh}{h} \times 100 \dots \dots (2) \quad \text{Ans} = 6\% \dots \dots (1)$$

Module-III

$$15 \text{ a) } \text{Region of integration} \dots \dots (1) \iint_R y \, dx \, dy = \int_0^4 \int_{\frac{y^2}{4}}^{2\sqrt{y}} y \, dx \, dy \dots (2)$$

$$\int_0^4 (2y^{\frac{3}{2}} - \frac{y^3}{4}) \, dy \dots \dots (3) = \frac{48}{5} \dots \dots (1)$$

OR

$$\text{Region of integration} \dots \dots (1) \iint_R y \, dx \, dy = \int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} y \, dy \, dx \dots (2)$$

$$\int_0^4 (4x - \frac{x^4}{16}) \, dx \dots \dots (3) = \frac{48}{5} \dots \dots (1)$$

$$b) \text{Region of integration} \dots \dots (1)$$

$$\text{Area} = \iint_R dx \, dy \dots (1) \int_0^4 \int_{x^2/2}^{2x} dy \, dx \dots (2) \int_0^4 (2x - \frac{x^2}{2}) \, dx \dots (1) = \frac{16}{3} \dots (2)$$

OR

$$\text{Region of integration} \dots \dots (1)$$

$$\text{Area} = \iint_R dx \, dy \dots (1) \int_0^8 \int_{y/2}^{8\sqrt{2y}} dx \, dy \dots (2) \int_0^8 (\sqrt{2y} - \frac{y}{2}) \, dy \dots (1) = \frac{16}{3} \dots (2)$$

$$16 \text{ a) } \text{Region of integration} \dots \dots (1)$$

$$\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy = \int_0^1 \int_0^{2x} e^{x^2} \, dy \, dx \dots \dots (2) \int_0^1 e^{x^2} 2x \, dx \dots \dots (2) \quad e-1 \dots \dots (2)$$

$$b) \quad V = \iiint_G dV \dots (1) = \iint_R \int_1^{5-x} dz \, dy \, dx \dots (2)$$

$$\int_0^{2\pi} \int_0^3 (4 - r \cos \theta) r \, dr \, d\theta \dots (2) \int_0^{2\pi} (18 - 9 \cos \theta) \, d\theta \dots (1) = 36\pi \dots \dots (1)$$

OR

$$V = \iiint_G dV \dots (1) = \iint_R \int_1^{5-x} dz \, dy \, dx \dots (2)$$

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$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x) dy dx \dots \dots \dots (2) = 36\pi \dots \dots \dots (2)$$

Module-IV

17 a) $u_k = \frac{1.2.3 \dots k}{1.3.5 \dots (2k-1)} \dots \dots \dots (2)$ (7)

$$u_{k+1} = \frac{(k+1)!}{1.3.5 \dots (2k-1)(2k+1)} \dots \dots (1) \quad \rho = \lim_{k \rightarrow \infty} \frac{k+1}{2k+1} = \frac{1}{2} < 1 \dots \dots (3)$$

Hence converges $\dots \dots (1)$

b) (7)

$$|U_k| = \left| \frac{1}{\sqrt{k(1+k)}} \right| \dots \dots (1) \quad \sum_{k=1}^{\infty} V_k = \sum_{k=1}^{\infty} \frac{1}{k} \text{ Divergent } \dots \dots (1) \quad \lim_{k \rightarrow \infty} \frac{U_k}{V_k} = 1 \dots \dots (1)$$

Not Absolute Convergent $\dots \dots (1)$

$$U_1 > U_2 > \dots \dots (1) \quad \lim_{k \rightarrow \infty} U_k = 0 \dots \dots (1) \text{ conditionally convergent } \dots \dots (1)$$

18 a) $U_{k+1} = \frac{x^{k+1}}{(k+1)(k+2)} \dots \dots (1) \quad \lim_{k \rightarrow \infty} \frac{U_{k+1}}{U_k} = x \dots \dots (1)$ (7)

$x < 1$ Convergent, $x > 1$ Divergent $x = 1$ test fails $\dots \dots (1)$

$$\text{If } x=1 \quad U_k = \frac{1}{(k)(k+1)} \dots \dots (1) \quad \sum_{k=1}^{\infty} V_k = \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ convergent } \dots \dots (1) \quad \lim_{k \rightarrow \infty} \frac{U_k}{V_k} = 1 \neq 0 \dots \dots (1)$$

Convergent $\dots \dots (1)$

OR

$$\lim_{k \rightarrow \infty} \left| \frac{U_{k+1}}{U_k} \right| = |x| \dots \dots (1) \quad -1 < x < 1 \text{ series converges } \dots \dots (1)$$

$x = -1$, U_k decreases & $\lim U_k = 0$, so it converge at $x = -1$ $\dots \dots (1)$

$$\text{If } x=1 \quad U_k = \frac{1}{(k)(k+1)} \dots \dots (1) \quad \sum_{k=1}^{\infty} V_k = \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ convergent } \dots \dots (1) \quad \lim_{k \rightarrow \infty} \frac{U_k}{V_k} = 1 \neq 0 \dots \dots (1)$$

Convergent $\dots \dots (1)$

b) $U_{k+1} = \frac{(k+2)!}{4!(k+1)!4^{k+1}} \dots \dots (2) \quad \lim_{k \rightarrow \infty} \frac{U_{k+1}}{U_k} = 1/4 < 1 \dots \dots (4) \text{ Convergent } \dots \dots (1)$ (7)

- 19 a) (If the answer is correct without writing the formula give full mark)

(7)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + b_n \sin n\pi x \dots\dots(1)$$

$$\text{Formula } a_0 = 2 \int_0^1 x dx = 1 \dots (1+1) \quad \text{Formula } a_n = 2 \int_0^1 x \cos n\pi x dx = \frac{2((-1)^n - 1)}{n^2 \pi^2} \dots\dots(1+1)$$

$$b_n = 0 \dots\dots(1)$$

Deduction---(1)

OR

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\pi x + b_n \sin n\pi x \dots\dots(1)$$

$$\text{Formula } a_0 = \int_0^1 x dx = \frac{1}{2} \dots (1+1) \quad \text{Formula } a_n = 2 \int_0^1 x \cos n\pi x dx = \frac{2((-1)^n - 1)}{n^2 \pi^2} \dots\dots(1+1)$$

$$b_n = 0 \dots\dots(1)$$

Deduction---(1)

- b) (If the answer is correct without writing the formula give full mark.)

(7)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \dots\dots(2)$$

$$\text{Formula, } a_0 = \frac{2}{L} \int_0^L f(x) dx = \frac{kL}{2} \dots (1+1)$$

$$\text{Formula } a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2kL}{n^2 \pi^2} \left(2 \cos \frac{n\pi}{2} - 1 - (-1)^n \right) \dots\dots(1+2)$$

OR

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \dots\dots(2)$$

$$\text{Formula, } a_0 = \frac{1}{L} \int_0^L f(x) dx = \frac{kL}{4} \dots (1+1)$$

$$\text{Formula } a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2kL}{n^2 \pi^2} \left(2 \cos \frac{n\pi}{2} - 1 - (-1)^n \right) \dots\dots(1+2)$$

- 20 a) (If the answer is correct without writing the formula give full mark)

(7)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \dots\dots(1)$$

$$\text{Formula, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{3} \dots\dots(1+1)$$

$$\text{Formula, } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{2(-1)^n}{n^2} \dots\dots(1+1)$$

$$\text{Formula, } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \sin nx dx = \frac{2((-1)^n - 1)}{\pi n^3} + \frac{\pi(-1)^{n+1}}{n} \dots\dots(1+1)$$

OR

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\pi x + b_n \sin n\pi x \dots\dots(1)$$

$$\text{Formula, } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{6} \dots\dots(1+1)$$

$$\text{Formula, } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{2(-1)^n}{n^2} \dots\dots(1+1)$$

$$\text{Formula, } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \sin nx dx = \frac{2((-1)^n - 1)}{\pi n^3} + \frac{\pi(-1)^{n+1}}{n} \dots\dots$$

(7)

b)

(If the answer is correct without writing the formula give full mark)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \dots\dots(2)$$

$$\text{Formula, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2\pi^2}{3} \dots\dots\dots(1+1)$$

$$\text{Formula, } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{4(-1)^n}{n^2} \dots\dots\dots(1+1)$$

$$\text{Formula, } b_n = 0 \dots\dots\dots(1)$$

OR

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \dots\dots\dots(2)$$

$$\text{Formula, } a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{\pi^2}{3} \dots\dots\dots(1+1)$$

$$\text{Formula, } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{4(-1)^n}{n^2} \dots\dots\dots(1+1)$$

$$\text{Formula, } b_n = 0 \dots\dots\dots(1)$$
